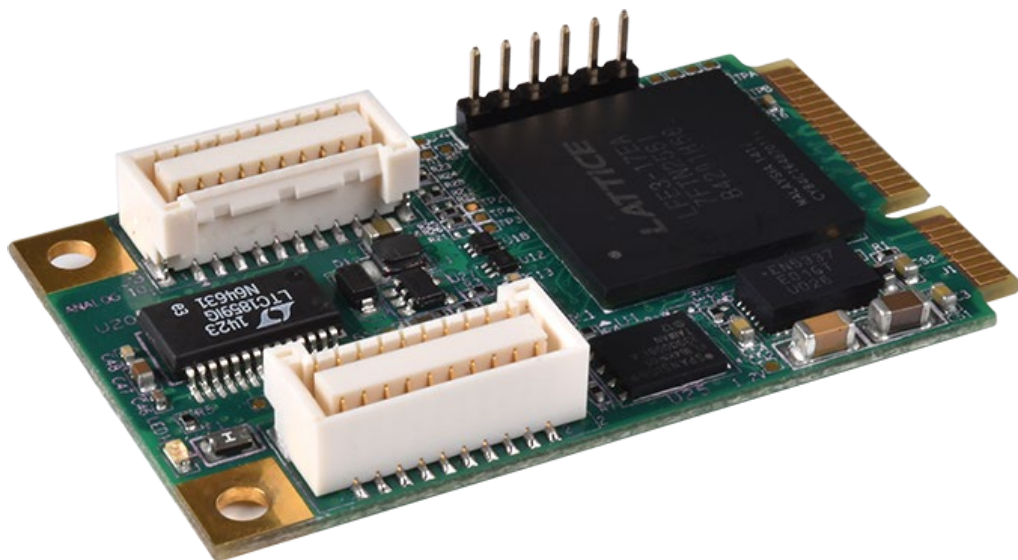


IO183

Analog I/O Module

Hardware Reference Manual



v1.0

speedgoat
real-time simulation and testing

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CONTENTS

Document Version History	3
1 Technical Information	4
2 Technical Specifications	5
3 Handling and Operating Instructions	6
3.1. ESD Protection	6
3.2. I/O Interface Installation	6
3.3. Assembly Recommendations	6
4 Functional Description	7
4.1. Analog-to-Digital Circuit	7
4.1.1. A/D Circuit Overview	7
4.1.2. A/D Operation	7
4.1.3. A/D Resolution	8
4.1.4. Input Range Selection	8
4.1.5. Measurement Accuracy and Calibration	8
4.1.6. Input Impedance	8
4.2. Digital-to-Analog Circuit	9
4.2.1. Overview	9
4.2.2. D/A Resolution	9
4.2.3. Output Range Selection	9
4.2.4. Output Accuracy and Calibration	9
4.3. Digital I/O Features	10
4.3.1. Overview	10
5 I/O Electrical Interface	11
5.1. A/D Circuit	11
5.2. D/A Circuit	11
5.3. Digital I/O	11
5.4. Data Acquisition Circuit Description	11
6 I/O Connector	13
6.1. Pin Mapping	13
7 Legal	14
7.1. Limited Warranty	14
7.2. Return	15
7.3. Systems Software Maintenance and Support Services	15
7.4. Use of Speedgoat Software Including Tools and Drivers	15
8 Contact Information	16

Document Version History

Version Number	Document Action	Author	Date
1.0	Initial release	BS	01 October 2019

1 Technical Information

Description

The IO183 I/O module is a fast, simultaneous-sampling, 16-bit analog input and output module with Simulink® driver blocks.

The module features eight single-ended (or four differential) analog inputs, four single-ended analog outputs and 14 digital I/O lines.

The IO183 I/O module can be connected by trigger lines to additional I/O modules to synchronously sample and convert analog input or output channels across the different modules.

This I/O module is ideal for closed-loop controls and hardware-in-the-loop simulations using MATLAB® and Simulink.

Features

Analog Inputs

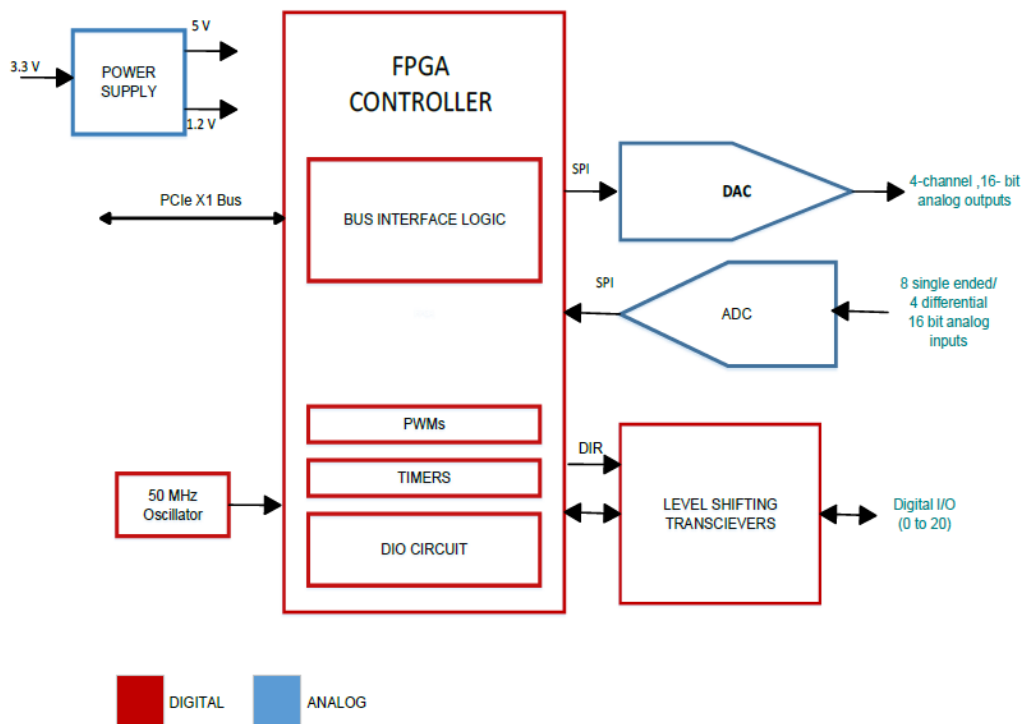
- 8 single-ended or 4 differential inputs
- 16-bit resolution
- Software selectable ± 10 V, ± 5 V, 0-10 V, 0-5 V input range (one setting for all channels)

Analog Outputs

- 4 single-ended outputs
- 16-bit resolution Software selectable 0-2.5 V, 0-5 V output range (one setting for all channels)

Digital I/O

- 14 digital I/O lines individually configurable as input or output
- Software selectable pull up/down resistors



IO183 I/O module block diagram

2 Technical Specifications

Physical	
Form factor	mPCIe
Power requirements	0.462 W @ 3.3 V DC
Bus	PCI Express x1
Dimensions	50.95 mm x 30 mm
Weight	8.5 g
Connectors	Digital I/O: 17-pin M12 male Cable connector: M12 17-pos female connector Analog I/O: 17-pin M12 female Cable connector: M12 17-pos male connector
Terminal Board	Terminal Board M12 17-pin for digital and analog signals Dimensions: 214 x 90 x 47 mm
Environmental	
Operating temperature	-40 °C to +85 °C
Relative humidity	5 to 95 %, non-condensing
Analog Input	
Number of inputs	8 single ended or 4 differential
A/D resolution	16 bits
Sample rate	100 kHz maximum aggregate
Voltage ranges	Software selectable ± 10 V, ± 5 V, 0-10 V, 0-5 V (one setting for all channels)
Input over voltage protection	Up to ± 25 V
Input impedance	31 k Ω typical (for bipolar input ranges), 42 k Ω typical (for unipolar input ranges)
Analog input error	± 28 LSB (± 0.04 %) max zero scale error (no offset adjustment)
Analog Output	
Number of outputs	4 single ended
D/A resolution	16 bits
Voltage ranges	Software selectable 0-2.5 V, 0-5 V output range (one setting for all channels)
Output updates	Simultaneous or individual channel
Output current	Up to 5 mA per channel (1 k Ω minimum load)
Output drift	Low drift 2 ppm/°C internal reference
Output error	± 1.5 mV (± 0.03 %) max zero scale error (no offset adjustment)
Digital I/O	
Configuration	14 lines buffered, individual direction control and protection
Output voltage	+3.3 V
Pull-up/pull-down	Software selectable

3 Handling and Operating Instructions

3.1. ESD Protection



The module is sensitive to static electricity. Appropriate care must be taken when packing, unpacking and handling the module.

3.2. I/O Interface Installation



Signals are available on both the front and rear connectors. Only one connector can be used/installed at a time.

3.3. Assembly Recommendations



When removing the module from the carrier board, ensure mechanical stress is kept as low as possible.

4 Functional Description

4.1. Analog-to-Digital Circuit

4.1.1. A/D Circuit Overview

The IO183 I/O module implements an analog input circuit based on the LTC1859 16-bit A/D converter, or ADC. The LTC1859 can operate at up to 100,000 samples per second in total, either on a single channel or on any number of channels. The ADC supports programmable input ranges of 0-5 V, 0-10 V, +/-5 V, and +/-10 V. The input range can be changed on a sample-by-sample or channel-by-channel basis.

The ADC includes a built-in channel multiplexor which can treat the eight inputs as either eight single-ended channels or four differential channels. In single-ended mode, each channel is measured with respect to analog ground, meaning the input signal must be connected between the input pin and the analog ground pin on the I/O connector. In differential mode, the voltage is measured as the difference between the two input pins, and the analog ground pin is not part of the measurement. However, both the high and low side of the input signal pair must remain within the "common mode voltage" of the A/D at all times to ensure accurate readings.

4.1.2. A/D Operation

In essence, an A/D converter converts an analog voltage to a number that indicates the ratio between that voltage and the full range of voltages that the A/D can measure. A 16-bit A/D converter means that all input voltages are represented by a 16-bit binary number ranging from 0 to 65535 ($2^{16}-1$), or binary 0000 0000 0000 0000 to 1111 1111 1111 1111. There are two groups of input ranges, unipolar and bipolar. Unipolar ranges consist of only positive voltages (0-5 V, 0-10 V), and bipolar ranges include both negative and positive voltages (+/-5V, +/-10V).

The tables below summarize the relationship between input voltage and A/D values for the general case. Note that the nominal upper limit of the input range (5.0000 V or 10.0000 V) is not achievable, since this voltage would require a 17-bit number (2^{16} or 1 0000 0000 0000 0000). V_{FS} means the full-scale input voltage, either 5 V or 10 V.

Bipolar Input Ranges

A/D Code	A/D Code Binary	Formula	+/-5 V Input Range	1/-10 V Input Range
-32768	1000 0000 0000 0000	$-V_{FS}$	-5.0000 V	-10.0000 V
-32767	1000 0000 0000 0001	$-V_{FS} + 1 \text{ LSB}$	-4.9998 V	-9.9997 V
-1	1111 1111 1111 1111	-1 LSB	-0.153 mV	-0.305 mV
0	0000 0000 0000 0000	0 V	0.0000 V	0.0000 V
1	0000 0000 0000 0001	+1 LSB	0.153 mV	.305 mV
32767	0111 1111 1111 1111	$V_{FS} - 1 \text{ LSB}$	4.9998 V	9.9997 V

Unipolar Input Ranges

A/D Code	A/D Code Binary	Formula	0-5 V Input Range	0-10 V Input Range
0	0000 0000 0000 0000	0 V	0.0000 V	0.0000 V
1	0000 0000 0000 0001	1 LSB ($V_{FS} / 65536$)	0.076 mV	0.153 mV
32767	0111 1111 1111 1111	$V_{FS} / 2 - 1 \text{ LSB}$	2.4999 V	4.9998 V
32768	1000 0000 0000 0000	$V_{FS} / 2$	2.5000 V	5.0000 V
32769	1000 0000 0000 0001	$V_{FS} / 2 + 1 \text{ LSB}$	2.5001 V	5.0002 V
65535	1111 1111 1111 1111	$V_{FS} - 1 \text{ LSB}$	4.9999 V	9.9998 V

4.1.3. A/D Resolution

The smallest change in input voltage that can be detected is $1/(2^{16})$, or $1/65536$, of the full-scale input range. This smallest change results in an increase or decrease of 1 in the A/D code, and is referred to as 1 LSB. The resolution is always 16 bits, but the value of 1 LSB will vary with the input range.

Polarity	Input Range	Resolution (1 LSB)
Bipolar	± 10 V	305 μ V
Bipolar	± 5 V	153 μ V
Unipolar	0 – 10 V	153 μ V
Unipolar	0 – 5 V	76 μ V

4.1.4. Input Range Selection

The input range of the A/D circuit is programmable in software and can be selected from the 4 values shown in the table above. The input range can be changed anytime, so that you can use different ranges for different input signals based on the best match. In general, you should select the highest gain (smallest input range) that allows the A/D converter to read the full range of voltages over which the input signals will vary. For example, if you have a signal that ranges from 0 V minimum to 3 V maximum, use the 0-5 V range for the best resolution. An input range that is too small causes the A/D converter to clip at either the high end or low end, and you will not be able to read the full range of voltages on your input signals.

4.1.5. Measurement Accuracy and Calibration

Although the A/D circuit has 16-bit resolution (meaning it can resolve input voltages to within $1/2^{16}$ or $1/65536$ of its input range), the measurement accuracy is not necessarily the same. All A/D circuits exhibit two inherent errors known as offset and gain errors. In addition all A/D circuits will experience some minor drift as the ambient temperature changes, as well as lesser drift over long periods of time.

To minimize the size and cost of the A/D circuit while still providing reasonable performance, the A/D circuit on the IO183 I/O module does not provide calibration features to eliminate these errors. Instead the circuit was designed to minimize the inherent offset, gain, and drift characteristics. The technical specifications table indicates the worst-case offset, gain, and temperature drift errors in the circuit.

If higher accuracy is required, a software or digital calibration technique can be implemented in your application. Measure two known voltages close to the lower and upper limits of the input range you are using. The ideal values are between 1 % and 5 % at the bottom end and 95 % and 99 % at the top end. Compare the actual A/D readings of these input voltages to the expected values, and then store the differences in a file. Then you can use these errors to apply an offset and gain factor to each A/D reading on the module permanently. Although this technique does not eliminate temperature and drift errors, it will eliminate the largest portion of the total error and provide more accurate measurements and better overall product performance.

4.1.6. Input Impedance

Another factor that can significantly affect A/D measurement accuracy is the ratio between the input impedance of the A/D circuit and the output impedance of the signal being measured. The voltage seen by the A/D converter is a simple resistor divider between the two impedances, with the input signal at the top of the divider and the A/D in the middle. The higher the output impedance of the input signal, the greater the error will be. Conversely, the higher the input impedance of the A/D circuit, the lower the error will be. The overall circuit input impedance is equal to the A/D converter's input impedance. For bipolar input ranges, the input impedance is 31 k Ω typical, and for unipolar input ranges the input impedance is 42 k Ω typical. For low errors, the formula for calculating the maximum output impedance of the input signal may be approximated as follows:

$$\text{Maximum input signal impedance} = \text{A/D impedance} \times \text{tolerance}$$

For an allowable 0.1 % error (66 LSB), the input signal should have an output impedance of 31 Ω max for bipolar input ranges and 42 Ω max for unipolar input ranges. Ideally the output impedance of the circuit should be less than 10 Ω .

4.2. Digital-to-Analog Circuit

4.2.1. Overview

The IO183 I/O module data acquisition circuit uses a 16-bit D/A converter. This means that all output voltages are defined by a 16-bit binary number, or D/A code, ranging from 0 to 65535 ($2^{16}-1$), or binary 0000 0000 0000 0000 to 1111 1111 1111 1111. The IO183 I/O module D/A circuit supports only unipolar output voltage ranges (positive voltages only). For a unipolar output range, the D/A codes are straight binary and will range from 0 (binary 0000 0000 0000 0000) for 0 V to 65535 (binary 1111 1111 1111 1111) for the top end of the range.

D/A Code	D/A Code Binary	Formula	0-5 V Output Range	0-2.5 V Output Range
0	0000 0000 0000 0000	0 V	0.0000 V	0.0000 V
1	0000 0000 0000 0001	1 LSB ($V_{FS} / 65536$)	0.0763 mV	0.0381 mV
32767	0111 1111 1111 1111	$V_{FS} / 2 - 1 \text{ LSB}$	2.4999 mV	1.24996 mV
32768	1000 0000 0000 0000	$V_{FS} / 2$	2.5000 V	1.25000 V
32769	1000 0000 0000 0001	$V_{FS} / 2 + 1 \text{ LSB}$	2.5001 V	1.25004 mV
65535	1111 1111 1111 1111	$V_{FS} - 1 \text{ LSB}$	4.9999 V	2.49996 V

4.2.2. D/A Resolution

The smallest change in output voltage that can be obtained is $1/(2^{16})$, or $1/65536$, of the full-scale output range. This smallest change results from an increase or decrease of 1 in the D/A code and is therefore referred to as 1 LSB (1 Least Significant Bit). The D/A resolution is always 16 bits, but the value of 1 LSB will vary with the output range.

Polarity	Output Range	Resolution (1 LSB)
Unipolar	0-5 V	76.3 μV
Unipolar	0-2.5 V	38.1 μV

4.2.3. Output Range Selection

The output range of the D/A circuit is programmable in software and can be selected from the 2 values shown in the table above. The output range can be changed anytime. However, unlike the A/D circuit, changing the output range affects all output channels simultaneously. In general, you should select the smallest output range that allows the D/A converter to cover the full range of output voltages you need for your application. For example, if you need the output to vary from 0 V minimum to 2 V maximum, use the 0-2.5 V output range for the best resolution. Note that the IO183 I/O module D/A circuit cannot output negative voltages.

4.2.4. Output Accuracy and Calibration

Although the D/A circuit has 16-bit resolution (meaning it can resolve output voltages to within $1/2^{16}$ or $1/65536$ of its output range), the output accuracy is not necessarily the same. As with A/D circuits described earlier, all D/A circuits exhibit two inherent errors known as offset and gain errors. In addition, all D/A circuits will experience some minor drift as the ambient temperature changes, as well as lesser drift over long periods of time.

To minimize the size and cost of the D/A circuit while still providing reasonable performance, the D/A circuit on the IO183 I/O module does not provide calibration features to eliminate these errors. Instead

the circuit was designed to minimize the inherent offset, gain, and drift characteristics. The technical specifications table indicates the worst-case offset, gain, and temperature drift errors in the circuit.

If higher accuracy is required, a software or digital calibration technique can be implemented in your application. Write the bottom and top output codes (0 and 65535) to the D/A circuit and measure the actual output voltages. Compare the actual voltages to the expected values, and then store the differences in a file. Then you can use these errors to apply an offset and gain factor to the D/A conversion formula for the module permanently. Although this technique does not eliminate temperature and drift errors, it will eliminate the largest portion of the total error and provide more accurate measurements and better overall product performance.

4.3. Digital I/O Features

4.3.1. Overview

The pins 1-16 are Digital I/O lines, where pin 16 can be optionally configured as a 3.3V source during assembly, prior to order. Pin 17 is the digital ground.

All lines operate with 3.3 V logic levels. They may be configured via software for 10 k Ω pull-up or pull-down resistors. The selected pull configuration is stored in non-volatile memory and is automatically recalled during power-up.

All digital I/O lines utilize 1-bit logic buffers (transceivers) to provide enhanced output current capability and protect the FPGA from faulty connections. The directions for all lines are individually configurable. For safety and to prevent glitches, on power-up or reset, all the lines reset to input mode and all the port data registers reset to 0.

5 I/O Electrical Interface

5.1. A/D Circuit

The module uses the Linear Technology LT1859 high-integration A/D converter chip for the A/D functionality and includes the following features:

- 16-bit A/D with 100 KHz sampling rate
- 8-channel single-ended/4-channel differential multiplexor
- Input protection up to ± 25 V
- Programmable input ranges: 0-10 V, 0-5 V, ± 10 V, ± 5 V
- Precision 2.5 V low-drift reference voltage

5.2. D/A Circuit

The module uses the Analog Devices AD5686R D/A converter chip for the D/A functionality and includes the following features:

- 4 channel 16-bit D/A
- Single-channel and simultaneous update modes
- Programmable output ranges: 0-2.5 V, 0-5 V
- Precision 2.5 V low-drift reference voltage

5.3. Digital I/O

The 14 digital I/O lines are provided by the FPGA and can operate in simple I/O mode in the form of 8-bit and 1-bit ports. All bits have independent 1-bit drivers with independent direction control.

The digital I/O output voltage is +3.3 V. The digital I/O lines can be software-configured for pull-up/-down resistors. All lines are configured together for up or down.

5.4. Data Acquisition Circuit Description

A/D Features

- 16 analog voltage inputs
- 16-bit resolution (1 part in 65536)
- Programmable input ranges: 0-5 V, 0-10 V, ± 5 V, ± 10 V
- Single-ended and differential input configuration options
- Precision, low-drift 2.5 V reference voltage
- 100 KHz maximum total A/D sample rate (all active channels combined)

D/A Features

- 4 analog voltage outputs
- 16-bit resolution (1 part in 65536)
- Single-channel and multi-channel simultaneous update modes
- Programmable output range: 0-5 V, 0-2.5 V
- 30 KHz update rate capability
- Waveform generator on 1 to 4 outputs with user-defined waveforms and 2048-sample waveform buffer

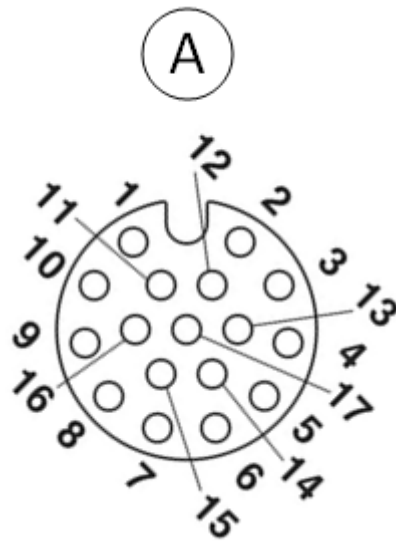
Digital I/O features

- 16 + 1 digital I/O lines and ground
- 3.3 V logic levels
- User-configurable 10 K pull-up/pull-down resistors
- Individually programmable direction for each line
- Buffers for protection and higher current drive

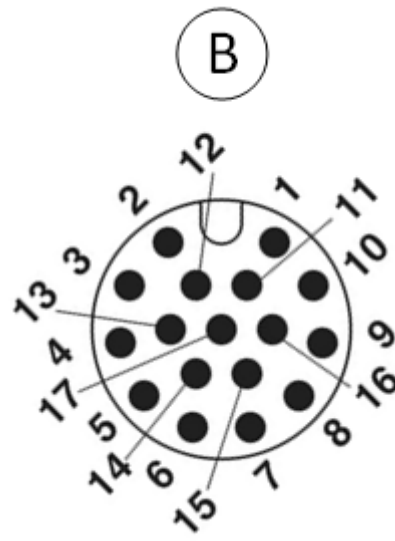
6 I/O Connector

6.1. Pin Mapping

Pin count	17 pins
Connector type	SACC-CI-M12MS-17CON-L180 (Phoenix Contact part no. 1442081)
Mating connector type	M12 17-pin female connector, A-coded



17-pin M12 connector (female)



17-pin M12 connector (male)

Connector 1 Signal				Connector 2 Signal
Single-Ended	Differential	Pin	Pin	
Analog Output 01	Analog Output 01	1	1	Digital I/O 01
Analog Output 02	Analog Output 02	2	2	Digital I/O 02
Analog Output 03	Analog Output 03	3	3	Digital I/O 03
Analog Output 04	Analog Output 04	4	4	Digital I/O 04
Analog Ground	Analog Ground	5	5	Digital I/O 05
Analog Ground	Analog Ground	6	6	Digital I/O 06
Analog Input 01	Analog Input 01	7	7	Digital I/O 07
Analog Input 02	Analog Input 02	8	8	Digital I/O 08
Analog Input 03	Analog Input 03	9	9	Digital I/O 09
Analog Input 04	Analog Input 04	10	10	Digital I/O 10
Analog Input 05	Analog Input 05	11	11	Digital I/O 11
Analog Input 06	Analog Input 06	12	12	Digital I/O 12
Analog Input 07	Analog Input 07	13	13	Digital I/O 13
Analog Input 08	Analog Input 08	14	14	Digital I/O 14
0 V	0 V	15	15	Digital I/O 15
+5 V	+5 V	16	16	Digital I/O 16 (+3.3V option)
Analog Ground	Analog Ground	17	17	Digital Ground

7 Legal

As used herein, the term "Seller" shall mean Speedgoat GmbH, and term "Buyer" shall mean the person, firm or corporation executing a purchase order for "goods", sold by Seller (hereinafter "Products").

7.1. Limited Warranty

Seller warrants that the Products delivered hereunder shall be free from defects in workmanship and material under normal use and wear in accordance with Seller's instructions and specifications for a period of twenty-four (24) months from date of delivery to the Buyer, including component parts of Products sold as spare, replacement, maintenance or storage parts, which are also warranted for twenty-four (24) months from date of delivery, provided, however, in either case, that notice of any such defect is provided to Seller within thirty (30) days of its discovery by the Buyer. THE WARRANTY SET FORTH IN THIS SECTION SHALL BE IN LIEU OF ALL OTHER WARRANTIES, AND ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTY OF MERCHANTABILITY, FITNESS FOR PARTICULAR PURPOSE, AND FITNESS FOR ORDINARY PURPOSE USED OR PURPOSE INTENDED, ARE EXCLUDED. IN NO EVENT SHALL SELLER, ITS EMPLOYEES OR SUPPLIERS BE LIABLE, EITHER DIRECTLY OR BY WAY OF INDEMNIFICATION, TO BUYER OR ANY THIRD PARTY FOR (A) AN AMOUNT EXCEEDING THE PURCHASE PRICE OF THE PRODUCT IN QUESTION AND (B) ANY PUNITIVE, EXEMPLARY, SPECIAL, INDIRECT OR CONSEQUENTIAL LOSSES, DAMAGES OR INJURIES regardless of whether such claim is based upon delays in delivery or payment, breach of warranty, breach of contract, strict liability, negligence, or any theory now known or hereinafter adopted by legislation or by any court. Neither Seller nor its affiliates shall be liable for any damage or loss to exposure of Products and/or their packaging to the elements (including but not limited to rain, snow, sleet, sun, wind, floods, etc.); chemicals, corrosive solvents or soils; unauthorized or improper use, maintenance, storage or repair; due to any failure to follow Seller's manuals, warnings, notices or instructions; or due to any malfeasance, recklessness or negligence by Buyer, any employee or costumer of Buyer or any other third party.

EXCLUSIVE REMEDY: In any event, the Buyer's exclusive remedy hereunder is limited to the furnishing of replacement parts on an exchange basis, or, at the option of Seller, to the repair or replacement of defective Products or replacement parts at one of Seller's locations, but in either case only if the defective Product or part has been submitted to Seller during the period of warranty. The Buyer accepts and acknowledges that the foregoing allocation of risk is reflected in the purchase price.

The parties further agree that if any portion of the foregoing exclusion of damages is held to be voidable or void by reason of public policy or unenforceable for any other reason whatsoever, all remaining portions of the foregoing exclusion shall continue in effect.

THE WARRANTY SET FORTH ABOVE DOES NOT EXTEND TO: Any systems which have been damaged or rendered defective as a result of accident, misuse, or abuse; by the use of parts not manufactured, authorized or sold by Seller; by modification or as a result of service by anyone other than Seller; systems not containing original components or original replacement of components; damage during shipment, unless due to incorrect packaging by Seller; systems which fail or are damaged after delivery due to shipment, handling, storage, operation, use or maintenance in manner or environment not conforming to any published instructions or specifications issued by Seller.

In-warranty repaired or replacement parts or Products are covered by warranty only for the remaining unexpired portion of the original warranty period applicable to the repaired or replaced parts or Products. In other words, repair or replacement of Products or parts under warranty does not extend the original warranty period.

Products that are no longer part of the regular sales offering are considered EOL (end-of-life) and are repaired on a best-effort basis.

EXTENDED HARDWARE WARRANTY SERVICE

Extended Hardware Warranty Service is available as an option and must be purchased at the time the Products are purchased for which the warranty shall be extended.

The **Level One Hardware Warranty Service** extends the standard 24-month warranty period by 12 months resulting in a 36-month warranty period.

The **Level Two Hardware Warranty Service** extends the standard 24-month warranty period by 36 months resulting in a 60-month warranty period.

Hardware warranty terms exceeding the 60-month range are available on request.

7.2. Return

Buyer shall not return any Product without Seller's prior written consent. An R.M.A. (Return Material Authorization) number issued by Seller must accompany all returned material. An RMA number can be obtained by contacting the Seller's support department (support@speedgoat.com).

Within Warranty, Products returned and needing corrective repair are serviced at no-charge in accordance with the terms of Seller's Warranty policy.

Repairs on out of Warranty Products are performed at Buyer's expense.

Please pack the returned Products in their original shipping cartons, or in equivalent strong protective shipping cartons. Shipping costs from Buyer to Seller associated with warranty repairs or replacements shall be borne by the Buyer. Shipping costs for the return of repaired goods from Seller to Buyer shall be borne by Seller.

7.3. Systems Software Maintenance and Support Services

Delivery of Seller systems and hardware/software components by default include 12 months (1 year) of Systems Software Maintenance and Support Services.

Subscription to Systems Software Maintenance and Support Services includes access to Seller tools and driver software compatible with future releases of MathWorks software and professional technical support by phone and e-mail.

Subscription to Systems Software Maintenance and Support Services does not include free updates of existing custom implementations (FPGA bitstreams).

Software Maintenance and Support Renewal

For uninterrupted Systems Software Maintenance and Support Services in subsequent years Buyer may opt to renew its subscription annually to maintain its investment. Reinstatement if elapsed is possible on request, but incurs back maintenance charges of up to 6 months. Staying subscribed is the most cost-effective way to access latest advances and technical support.

7.4. Use of Speedgoat Software Including Tools and Drivers

LEGAL INFORMATION ABOUT THE USE OF SPEEDGOAT TOOLS AND DRIVERS: Speedgoat tools and drivers are optimized for hardware purchased from Seller and may be used only in conjunction with the hardware (serial no.) for which the tools and drivers were purchased for. Access to the Speedgoat tools and drivers is only available if the target machine component has active subscription to Systems Software Maintenance and Support Services.

Terms and Conditions for software components are defined in the Speedgoat End-User License Agreement (EULA).

8 Contact Information

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